Krishna Mula

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Education

San Jose State University San Jose, CA
Master of Science in Computer Science *Aug 2025 - May 2027*
Coursework: Distributed Systems, Database Systems, Cloud Computing, Machine Learning, NoSQL Database Systems.

Experience

PwC
Software Engineer *Aug 2022 - Jul 2025*

- Architected **RAG pipeline** for IT security compliance, embedding documents into ChromaDB for semantic retrieval and integrating **LLM inference** for automated validation against regulatory standards, **reducing review time by 70%**.
- Engineered centralized IT security knowledge base by fine-tuning **Llama 2**, automating **80%** of manual source aggregation and standardizing threat intelligence retrieval **across 6 teams**.
- Redesigned RAG pipeline inference layer as async FastAPI microservices, handling document uploads, query routing to ChromaDB, and LLM inference responses, **reducing end-to-end latency by 30%**.
- Built ETL pipeline using PyRFC to extract SM20/SM21 security audit logs from **5 production SAP systems**, automating a process that previously required **2+ hours of manual export and formatting** per report.
- Automated security controls execution on client SAP systems by codifying **25+ controls** as Flask + MongoDB REST APIs, ingesting data directly from SAP HANA and validating against security specifications, **cutting audit effort by 60%**.
- Deployed serverless REST APIs on Azure Functions and AWS Lambda with pre-warm triggers, **serving compliance reports across 15+ systems**, optimizing for 20-minute twice-daily burst windows.

PwC
Software Engineering Intern *Feb 2022 - Aug 2022*

- Built Nivo.js dashboards tracking security control pass/fail rates, compliance coverage, and historical trends, enabling teams to monitor compliance posture and prioritize which controls to automate next.
- Automated deployments with **Azure CI/CD** pipelines, reducing release cycles from **2 days to 4.5 hours**.

Projects

Aether | *Python · TCP Sockets · FastAPI · Docker SDK · React · D3.js*

- Built a **distributed pub-sub messaging system** over raw TCP sockets with gossip-based message propagation across a multi-broker mesh, UUID deduplication, configurable fanout/TTL, and heartbeat-driven peer eviction.
- Implemented **Chandy-Lamport global snapshot algorithm** with snapshot replication to k peers, enabling a **hybrid broker failover system** that detects failures via health polling, queries surviving peers for snapshots, and either restores full broker state or redistributes orphaned subscribers to least-loaded brokers.
- Built a **FastAPI orchestration control plane** that manages broker, publisher, and subscriber containers dynamically via the Docker SDK, with pull-based subscriber reassignment, WebSocket event broadcasting, and a live React dashboard visualizing topology, metrics, and failover events in real time.
- Designed **subscriber autonomous recovery**: independent Ping/Pong failure detection, exponential backoff with jitter on reconnect, and a central assignment API so subscribers self-heal without coordinated restarts.

CuriosityAI | *Flask, Next.js, ChromaDB, PyTorch, vec2text*

- Built an AI research discovery tool that retrieves academic papers from user-provided keywords and identifies underexplored research areas by detecting sparse regions in the embedding space.
- Applied **Gaussian Mixture Models** on **768-dim** paper embeddings to map research density and surface low-coverage areas in semantic space.
- Developed **vec2text** pipeline to decode optimized embeddings into research proposals, refined with Claude API for coherence.
- Created interactive **3D research landscape visualization** using **UMAP reduction** and **HDBSCAN clustering** with Flask backend and Next.js frontend.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C, C++, HTML, CSS, Shell Scripting.
Frameworks: Flask, Node.js, FastAPI, Django, Express.js, React, Next.js.
Databases: PostgreSQL, MongoDB, Redis, SQLite, DynamoDB.
Tools & Cloud: Git, Docker, AWS (EC2, S3, Lambda, CloudFront, DynamoDB, RDS, Amplify), Azure Cloud (Azure Functions, Azure VM, Azure EntraID), GitHub Actions, Nginx, Firebase.